Ratios, Rates and Proportions

Concept

Ratios compare quantities. Proportions describe equal ratios.

Ratio order

Red:blue is different from red:total. Match units first.

Unit rate

Divide to find the amount per 1 unit.

Proportion

$a/b = c/d$ means $ad = bc$, with non-zero denominators.

Sharing

Add ratio parts → find one part → multiply each share.

Example 1

Share 96 stickers in the ratio 3:5.

Solution

1. Total parts $= 3 + 5 = 8$. One part $= 96/8 = 12$.

2. Shares $= 3(12) = 36$ and $5(12) = 60$ stickers.

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Example 2

Six notebooks cost $15. Find the cost of nine at the same unit price.

Solution

1. Unit price = 15/6 = $2.50 per notebook.

2. Nine notebooks cost 9 × 2.50 = $22.50.

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Practice | your turn

Show your steps. Include units and restrictions where needed.

1. Simplify 24:36:60.

2. Find the speed for 150 km in 2.5 h.

3. Share $84 in the ratio 2:5.

4. Solve $x/8 = 15/12$.

5. A delivery costs $6 plus $2 per km. Is cost proportional to distance? Explain.

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Practice answers | check after trying

1. Divide by 12: 2:3:5.

2. 150/2.5 = 60 km/h.

3. One part = 84/7 = 12; shares $24 and $60.

4. 12x = 120, so x = 10.

5. No. At zero distance the fee is $6, and the cost per kilometre changes with distance.